

LeadFinder Pro Suite

AI-Powered Business Lead Discovery & CRM Automation Platform

LeadFinder Pro Suite is a production-grade AI-powered lead generation and CRM automation system designed for modern business intelligence workflows. The platform automates multi-source business scraping, real-time lead management, CRM tracking, automation pipelines, and advanced analytics across targeted geographic areas.

1. Project Overview

LeadFinder Pro Suite is a scalable lead discovery ecosystem that automates the collection, validation, organization, and management of business leads from multiple online sources including Google Maps, JustDial, and IndiaMart.

The platform focuses on area-wise business intelligence by enabling users to search, automate, and analyze businesses using structured geographic hierarchies such as State → City → Area.

The system supports real-time scraping, CRM management, automation workflows, advanced analytics dashboards, and export systems for enterprise-grade lead management.

2. Core Features

Feature	Description
Multi-Source Scraping	Google Maps, JustDial, IndiaMart integration
Real-Time Leads	Instant database sync and frontend updates
CRM Pipeline	Lead stages with follow-up tracking
Automation Engine	Area-wise automated lead scraping
Analytics Dashboard	Live statistics and business insights
Export System	CSV and Excel export functionality
Pagination & Filters	Fast scalable lead browsing
Area Heatmap	Visual geographic business density

3. Technology Stack

Frontend technologies include Next.js 15, React 19, TypeScript, Tailwind CSS, React Query, Zustand, and Recharts for modern UI development and state management.

Backend services are built using Node.js, Express.js, TypeScript, MongoDB, and Playwright-based scraping engines with Chromium automation support.

The system architecture is designed to support scalable scraping, real-time APIs, asynchronous automation processing, and enterprise-level lead workflows.

4. Frontend Architecture

The frontend application is built using a modular architecture with reusable components, centralized state management, responsive layouts, and dynamic data fetching.

Major frontend pages include: • Dashboard • Search • Leads • CRM Pipeline • Analytics • Automation • Settings

The UI follows a professional off-white enterprise theme optimized for usability, readability, and large-scale lead management operations.

5. Backend Architecture

The backend follows a service-oriented architecture with dedicated modules for scraping, lead management, automation, analytics, CRM operations, and exports.

Core backend modules: • Scraper Engine • Lead Service • CRM Service • Automation Queue • Analytics Engine • Export Manager • Authentication Middleware

All APIs are optimized for real-time operations and database synchronization between frontend and backend systems.

6. Database Design

Collection	Purpose
users	Authentication and admin accounts
leads	Business lead storage
crmPipeline	Lead workflow stages
automationJobs	Automation task management
analytics	Aggregated reporting metrics

The MongoDB database structure is optimized for scalable lead storage, realtime updates, and fast pagination-based retrieval.

7. Search & Scraping Workflow

The Search module allows users to search businesses using: Business Type → State → City → Area → Sources

The scraping engine launches Playwright Chromium sessions and dynamically extracts: • Business Names • Websites • Phone Numbers • Addresses • Categories • Ratings • Reviews

Extracted data is validated, deduplicated, stored in MongoDB, and immediately displayed on the frontend in real time.

8. CRM & Automation System

The CRM Pipeline provides a drag-and-drop workflow system for lead management.

Lead stages include: • New Lead • Contacted • Interested • Not Interested • Follow-Up

The Automation Engine supports: • Area-wise automation • Resume from stopped state • Multi-business-type scraping • Real-time automation progress • Continuous lead saving

9. Analytics & Reporting

The Analytics module provides live business intelligence insights and graphical visualizations.

Implemented analytics: • Total Leads • Scraping Success Rate • Automation Metrics • Lead Distribution
• Area Heatmap • Business Density Graphs

The Area Heatmap visually displays business density using color-coded geographic intelligence for premium lead analysis workflows.

10. Security & Performance

The platform includes enterprise-grade optimizations and security controls.

Security Features: • JWT Authentication • Protected Routes • Admin-only Access • Secure API Middleware • Session Persistence

Performance Optimizations: • Parallel scraping • Queue-based automation • Fast pagination • Realtime caching • Batched database writes

11. Final Summary

LeadFinder Pro Suite is a complete AI-powered lead generation ecosystem capable of handling large-scale business discovery, CRM workflows, automation operations, analytics reporting, and geographic business intelligence.

The system is production-ready and designed for enterprise sales teams, agencies, business intelligence operations, and automated lead generation workflows.

Key strengths of the platform: • Real-time lead generation • Multi-source scraping • CRM automation • Geographic targeting • Enterprise analytics • High scalability

Prepared for Project Delivery & Client Demonstration